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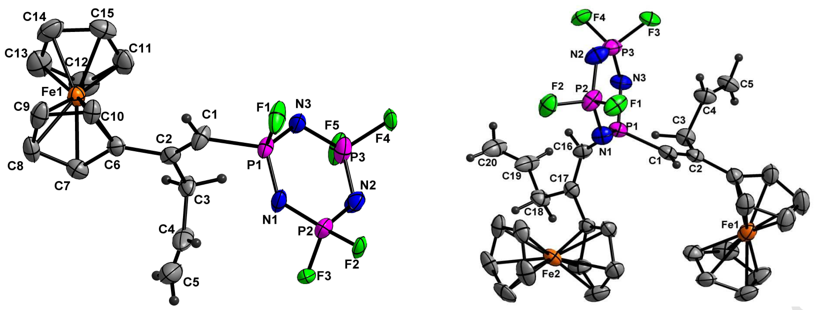
Graphical Abstract Synopsis

The first reactions of allylzinc bromide with mono and bis geminal ethynylferrocene derived fluorophosphazenes resulted in products having one to three allyl groups stereospecifically substituting on the alkyne units with no reaction on the P-F bonds of the fluorophosphazene.

Graphical Abstract Pictogram



Reactions of allylzinc bromide with ethynylferrocene derived fluorinated cyclophosphazenes

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ABSTRACT

The use of organozinc reagents in the synthesis of terminal alkene derived cyclic halogenated phosphazenes has been explored by the reactions of allylzinc bromide and ethynylferrocene derived fluorophosphazenes. The reaction of monoethynylferrocene derived fluorophosphazene, $[(FcC{\equiv}C)(F)PN](PNF_2)_2$ (Fc = ferrocenyl) with allylzinc bromide in 1:6 molar ratio resulted in the formation of $[FcC(CH_2CH{=}CH_2){=}CH(F)PN](PNF_2)_2$ **(1)** and $[FcC(CH_2CH{=}CH_2){=}C(CH_2CH{=}CH_2)(F)PN](PNF_2)_2$ **(2)** having allyl groups attached to the alkenyl carbon close to the ferrocene moiety. A similar reaction of geminal bisethynylferrocene derived fluorophosphazene, $[(FcC{\equiv}C)_2PN](PNF_2)_2$ with allylzinc bromide in 1:4 molar ratio resulted in the formation of $[\{FcC(CH_2CH{=}CH_2){=}CH\}_2PN](PNF_2)_2$ **(3)** while a reaction in 1:14 molar ratio resulted in the formation of compound **3** along with the unsymmetrically substituted triallyl derivative, $[\{FcC(CH_2CH{=}CH_2){=}C(CH_2CH{=}CH_2)\}\{FcC(CH_2CH{=}CH_2){=}CH\}PN](PNF_2)_2$ **(4)**. The new compounds **1-4** were characterized by ^{1}H, $^{13}C\{^{1}H\}$, $^{31}P\{^{1}H\}$, $^{19}F\{^{1}H\}$, IR and HRMS studies. Compounds **1** and **3** have also been structurally characterized by single crystal X-ray diffraction studies.

Keywords

Fluorophosphazene, Ethynylferrocene, Terminal alkenes, Allylzinc bromide, Stereospecific.

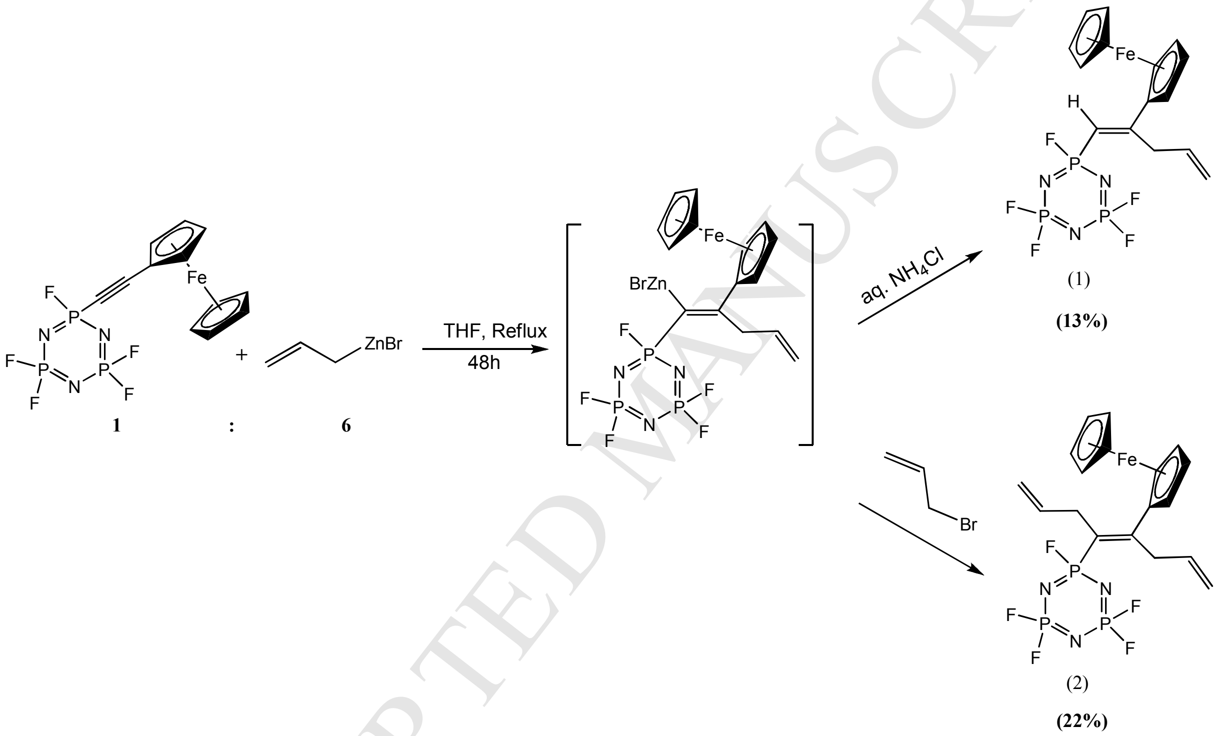
Introduction

Among inorganic heterocycles, cyclophosphazenes and their derivatives are unique since potentially useful and novel derivatives ranging from polymers, dendrimers, multidentate ligands, high energy materials and catalysts can be prepared using their unique structural framework and reactive phosphorus – halogen bonds [1-5]. One group of such cyclophosphazene derivatives, which are difficult to realize are those having terminal alkenyl groups bound to the phosphazene ring through direct P-C bonds. Previous studies by Allen and coworkers have resulted in a few alkene derived fluorophosphazenes obtained from the reaction of fluorinated cyclophosphazene with propenyllithium and (1-lithioalkoxy)ethylenes [6]. We have synthesized and carried out ring closing metathesis reactions on a range of terminal alkene derived cyclophosphazenes where the alkenyl units are bound to the phosphazene ring through P-O bonds [7a]. We have also recently observed the formation of alkenyl derived fluorophosphazenes by the controlled reduction of cyclophosphazene derived alkynes using Lindlar's catalyst [7b]. A range of synthetic methods using Grignard [8], organolithium [9] and organocopper [10] reagents as well as the Friedel-Crafts reaction [7a, 11] have been extensively utilized for making P-C bonded cyclophosphazene derivatives so far. The results from such studies, in addition to the expected derivatives have often resulted in unexpected phosphazene coupled products [12] as well as P-H bonded cyclophosphazenes [13].

Unlike Grignard and organolithium reagents, organozinc reagents are easy to prepare and handle and their reactions with unsymmetrical alkynes have been found to be stereo controlled. In general, allylzinc bromide has been reported to be a reagent of choice to introduce terminal alkenyl groups on unsymmetrically substituted alkynes with excellent stereo control [14]. It is interesting to note that the utility of organozinc reagents in making useful derivatives of cyclophosphazenes have not been reported so far. A recent report from our laboratory has described the reactions of ethynylphosphazene derived cyclophosphazenes with cobalt half-sandwich compounds as well as organic azides [9c]. Herein we report the first reactions of allylzinc bromide with mono and geminally disubstituted ethynylferrocene derived fluorophosphazenes which have resulted in a range of new alkenyl derived cyclophosphazenes having one or more allyl groups added to the alkyne unit regioselectively. The new compounds have been characterized by spectral and structural studies and the nature of products obtained in these reactions compared with non-phosphazenyl analogues.

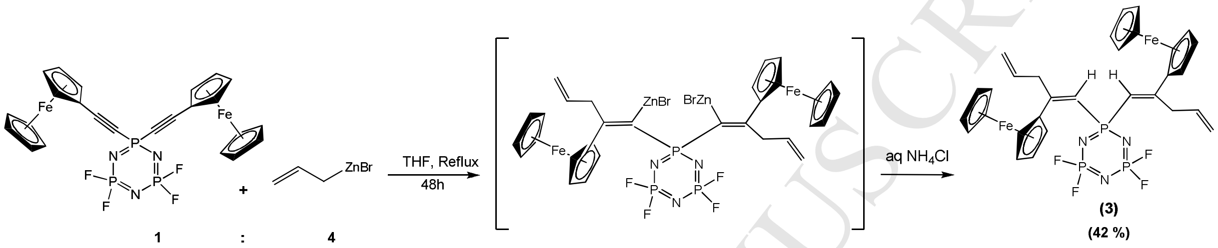
Results and discussion

The reactions of $(FcC{\equiv}C)P_3N_3F_5$ (Fc = ferrocenyl) with six equivalents allylzinc bromide resulted in $[FcC(CH_2CH{=}CH_2){=}CH(F)PN](PNF_2)_2$, **1** having mono substitution of allyl group on the alkynyl carbon which is directly connected to the electron donating ferrocene group in a *trans* orientation along with $[FcC(CH_2CH{=}CH_2){=}C(CH_2CH{=}CH_2)(F)PN](PNF_2)_2$, **2** with disubstitution on the alkynyl carbons also in a *trans* manner (Scheme 1). The reaction when attempted in lower stoichiometries resulted in poor yields of the same products. Both the products **1** and **2** goes through the same intermediate having ZnBr substituted on the ethynyl carbon adjacent to the electron withdrawing fluorophosphazene moiety. Compound **1** was formed upon hydrolysis of the intermediate with saturated ammonium chloride while **2** was formed possibly by the reaction of the same intermediate with excess of allyl bromide present in the reaction medium. A search of the literature indicates that this second reaction normally requires presence of some Ni [14b] and Pd [14e] catalysts or copper reagents [14a, 14c]. In the present study, possibility of zinc ions catalyzing this reaction cannot be ruled out. It was also of interest to note that unlike the reactions of Grignard and organolithium reagents, where reaction can take place on reactive P-F bonds, no phosphorous substituted product was obtained in our reactions.



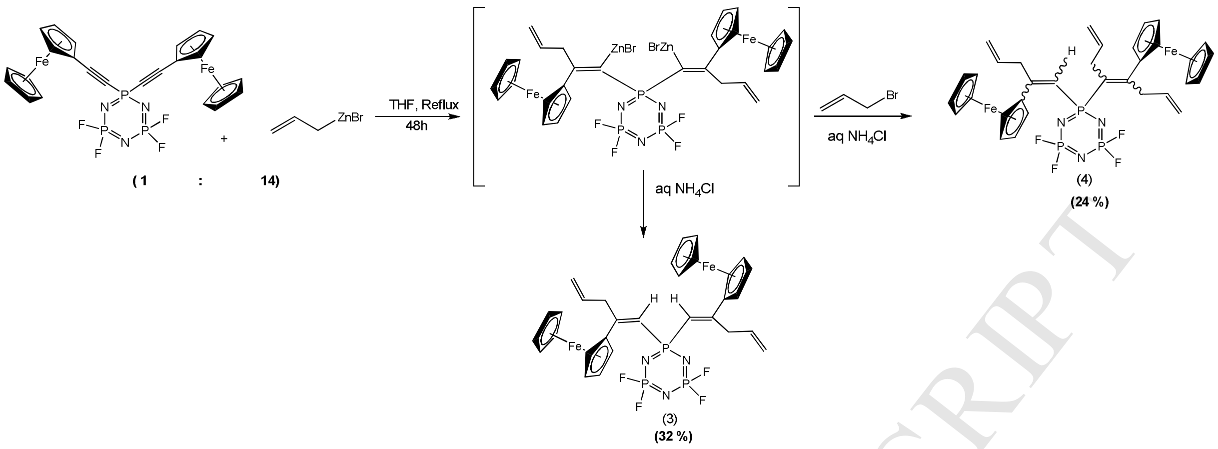
Scheme 1. Reactions of (FcC≡C)$P_3N_3F_5$ with in situ prepared allylzinc bromide in 1:6 molar ratio.

A similar reaction of geminal bisethynylferrocene derived fluorophosphazene, $[(FcC{\equiv}C)_2PN](F_2PN)_2$ with four equivalents of allylzinc bromide resulted in $[\{FcC(CH_2CH{=}CH_2){=}CH\}_2PN](PNF_2)_2$, **3** as the major product with substitution of allyl group on the same phosphazene bound alkynyl carbons of two different ferrocene bound ethynyl units (Scheme 2). X-ray crystallographic study revealed that unlike compound **1,** where the substitution was exclusively in the *trans* orientation with respect to the phosphazene and ferrocene moieties, in compound **3** one of the allyl groups was found to be *trans* oriented while the other was *cis* oriented.



Scheme 2. Reactions of $[(FcC{\equiv}C)_2PN](F_2PN)_2$ with allylzinc bromide in 1:4 molar ratio.

Using a large excess of allylzinc bromide (1:14 molar ratio) in the same reaction resulted in the formation of **3** along with an unsymmetrically trisubstituted compound $[\{FcC(CH_2CH{=}CH_2){=}C(CH_2CH{=}CH_2)\}\text{-}\{FcC(CH_2CH{=}CH_2){=}CH\}PN](PNF_2)_2$, **4** as a minor product, where one alkynyl group was disubstituted with the allyl group and the other alkynyl unit mono substituted with the same (Scheme 3). Compound **3** was formed on hydrolysis of the first intermediate with saturated NH_4Cl, whereas compound **4** was formed possibly from the reaction of the first intermediate with more of allyl bromide and hydrolysis with saturated NH_4Cl. Our attempts to prepare a tetraallyl substituted product by carrying out the same reaction in a higher boiling solvent (toluene) resulted in decomposition products.



Scheme 3. Reactions of [(FcC≡C)$_2$PN](F$_2$PN)$_2$ with allylzinc bromide in 1:14 molar ratio.

The reactions carried out in the present study were found to proceed in a similar manner as reported in the case of alkynyl sulfone with allylzinc bromide [14] where the first substitution was on the electron rich carbon as in our case also, but the stereochemistry of the product was *cis* in their case whereas in our case it was found to be in the *trans* orientation. Also, for getting the disubstituted product, Negishi Reagent, $NiCl_2(PPh_3)_2$ was used, whereas in our case diallyl substitution was achieved simply by the use of excess allyl reagent. Both **1** and **3** were obtained as dark red crystals under cold conditions from a mixture of hexane, ethyl acetate and toluene. Compounds **1** and **2** were found to be relatively unstable when compared to **3** and **4** and decomposed rapidly in air. Column chromatography was not found to be very useful for separation of compounds **1**-**4** because of very close R_f values and as the reaction always resulted in a mixture of products, some in un-isolable amounts. After removing the excess allylzinc bromide and decomposition products by flash chromatography over silica gel, preparative TLC in hexane-ethyl acetate (0.5:10) was carried out with the mixture to get the major compounds in the pure form. Based on our reaction conditions, we were able to get a maximum of two substitutions on mono alkynyl phosphazenes and three substitutions on geminal alkynyl phosphazenes. The effect of stoichiometry on the reactions were explored by carrying out the reactions in 1:4, 1:6, 1:10, 1: 12 and 1:14 molar ratios to maximize yields and to see the possibility of new product formation, if any. A reaction of $[(FcC{\equiv}C)_2PN](F_2PN)_2$ with allylzinc bromide in 1:14 molar ratio in the presence of Negishi reagent, $NiCl_2(PPh_3)_2$ was found to result in many products which were not separable either by column of preparative thin layer chromatography. The new fluorophosphazene derivatives having multiple terminal alkene units at its periphery are interesting precursors and provide scope for carrying out a variety of terminal alkene based reactions such as olefin metathesis, polymerization and alkene oxidation as well as reactions targeting the P-F bonds of the cyclophosphazene units.

Molecular structures of compounds 1 and 3

The crystal structure of compounds **1** and **3** are given in Figures 1 and 2, and the bond length and bond angles are given in Table 1. Crystallographic data and data collection parameters are given in Table 2. Suitable red colored crystals of compound **1** and **3** for X- ray study were obtained by slow evaporation of the compounds from a toluene/dichloromethane mixture. Compound **1** was found to crystallize in the monoclinic system whereas compound **3** crystallized in the orthorhombic crystal system. The phosphazene ring was found to be almost planar in both **1** and **3** as the substituted phosphorous atom and the opposite ring atom does not deviate significantly from the mean plane defined by the other four atoms of the phosphazene ring. X-ray structure of compound **1** shows that the allyl group is attached to the ethynyl carbon close to the ferrocene moiety in *trans* orientation. The structure of the geminally disubstituted compound **3** also shows that both the allyl groups are attached to the ethynyl carbon close to the ferrocene moiety. Interestingly one of the allyl groups is attached in the *trans* manner while the other allyl group is attached in the *cis* manner. The two ferrocene units of compound **3** were found to be nearly perpendicular to each other with the substituted Cp rings making an angle of 84.88(2)^o between them. The P-N bond distances involving the substituted phosphorous atoms were found to be in the range of 1.551(6) - 1.623(4) Å, which were longer than the other P-N bonds for both compounds **1** and **3**. Similarly, N(1)-P(1)-N(1) angles in compound **1** and **3** were found to be narrower than the other N-P-N angles of these compounds and the variation was found to be more noticeable in the geminal compound **3**. The P-C bond distances in compounds **1** and **3** were in the range of 1.748(5) - 1.760(5) Å, which is longer than those reported for their respective mono and geminal ethynylferrocene substituted cyclophosphazene derivatives [9c]. The C=C bond directly attached to the P atom was found to be longer [1.315(7) - 1.343(6) Å] than the other C=C bonds (of the allyl group) 1.057(1) - 1.294(7) Å.

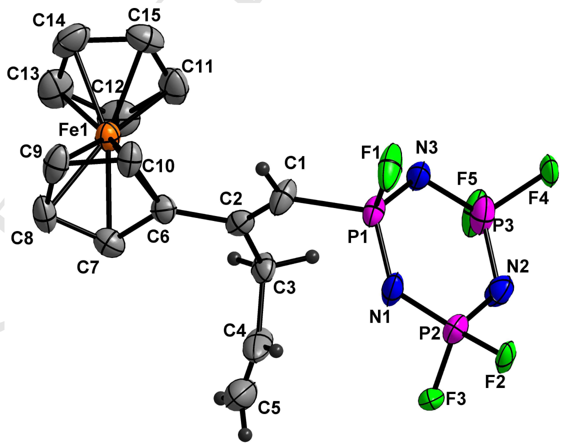


Fig. 1. ORTEP diagram of compound **1** with thermal ellipsoids at the 30% probability level. Ferrocene hydrogen atoms have been omitted for clarity.

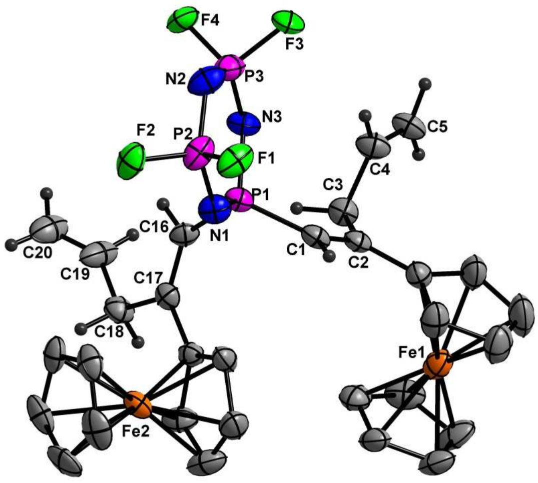


Fig. 2. ORTEP diagram of compound **3** with thermal ellipsoids at the 30% probability level. Ferrocene hydrogen atoms have been omitted for clarity

Table 1: Selected bond lengths and bond angles of compounds **1** and **3**

Bond Lengths (Å)			Bond Angles(deg)		
	Compound **1**	Compound **3**		Compound **1**	Compound **3**
P(1)-N(1)	1.551(6)	1.618(4)	N(1)-P(1)-N(3)	116.1(3)	112.6(2)
P(1)-N(3)	1.571(6)	1.623(4)	N(3)-P(3)-N(2)	120.3(3)	121.2(2)
P(2)-N(2)	1.553(6)	1.569(5)	N(1)-P(1)-C(1)	112.5(3)	105.0(2)
P(1)-F(1)	1.538(4)	1.521(3)	C(1)-C(2)-C(3)	123.9(5)	121.7(4)
P(1)-C(1)	1.748(5)	1.760(5)	C(2)-C(3)-C(4)	109.5(7)	112.7(4)
C(1)-C(2)	1.315(7)	1.343(6)	C(3)-C(4)-C(5)	142.0(2)	127.3(5)
C(2)-C(3)	1.492(8)	1.509(6)	C(1)-C(2)-C(6)	120.8(5)	120.2(4)
C(3)-C(4)	1.505(1)	1.501(6)	C(3)-C(2)-C(6)	115.3(5)	118.1(4)
C(4)-C(5)	1.057(1)	1.294(7)	C(7)-C(6)-C(10)	107.0(5)	106.6(5)
C(2)-C(6)	1.470(7)	1.452(6)			

Table 2 X- Ray crystal structure parameters of compounds **1** and **3**.

Parameters	**1**	**3**
Formula	$C_{15}H_{15}F_5FeN_3P_3$	$C_{30}H_{30}F_4Fe_2N_3P_3$
MW	481.06	713.18
cryst syst	monoclinic	othorhombic
Space group	$P2_1/n$	$P\,2_1\,2_1\,2_1$
a /Å	10.638(10)	9.906(2)
b /Å	7.617(7)	17.382(4)
c /Å	23.44(2)	17.970(4)
α (°)	90.00	90.00
β (°)	95.71(2)	90.00
γ (°)	90.00	90.00
V /Å^3	1890.0(3)	3094.2(1)
Z	4	4
T /K	298(2)	298(2)
λ /Å	0.71073	0.71073
ρ_c (g/cm3)	1.691	1.531
μ (mm-1)	1.104	1.143
Goodness of fit	0.814	0.990
θ range	1.75-25.00	1.63-25.00
Total reflections	8836	29850
Unique reflections	3319	5453
Ovserved data[$I > 2\sigma(I)$]	1521	3859
R_{int}	0.1247	0.0813
R_1, wR_2 [($I > 2\sigma(I)$]	0.0599, 0.1012	0.0495, 0.0918
R_1, wR_2 (all data)	0.1360, 0.1175	0.0806, 0.1013

$R_1 = \Sigma|F_o| - |F_c||/\Sigma|F_o|$; $wR_2 = \Sigma\,(|F_o|^2 - |F_c|^2)^2]\}^{½\,m}$

Spectral Studies

The ^{1}H-NMR chemical shifts for the cyclopentadienyl units of compounds **1-4** were observed in the range 4.13-4.62 ppm. For compound **1,** the chemical shifts of the alkene protons directly connected to the phosphorous (P-CH=) and the allyl proton CH= were found to be overlapping giving a complicated multiplet in the 5.80-6.09 ppm region whereas the same protons for compound **2** were observed around 5.83 ppm. The terminal $=CH_2$ protons are observed as a multiplet around 5.19 ppm for compound **1** whereas the same for compound **2** were observed around 5.09 ppm. For compounds **3** and **4,** the P-CH= protons came in the region of 6.07-6.10 ppm and the terminal allyl $=CH_2$ protons came in the region 5.10-5.19. The CH= protons for **3** came as multiplet in the region 3.01-5.90 ppm whereas the same for **4** was observed at 5.30 ppm. The $–CH_2$- protons for all the compounds appeared in the shielding region between 3.13 to 3.59 ppm. ^{13}C-NMR showed that on conversion of ethynyl to alkenyl units, signals corresponding to the ferrocenyl group get shielded. For example ferrocene cyclopentadienyl carbon signals change from 72.95 - 70.66 ppm in $[FcC{\equiv}C(F)PN](PNF_2)_2$ to 70.42 - 69.45 ppm in **1** and to 70.79 - 69.63 in **2**. In compounds **1** - **4**, signals corresponding to the alkene carbons of the allyl groups were found to be in the range of 116.29 - 118.75 ppm and the signals corresponding to the alkene directly bound to the P atoms were observed in the range of 134.08 - 136.97 ppm.

The ^{31}P-NMR spectrum for compounds **1** and **2** were observed as doublet multiplet for the PF moiety (at 36.36 and 35.08 ppm, respectively), and triplet multiplet for the PF_2 moiety (at 7.80 and 7.68 ppm, respectively) whereas the same for the starting material $(FcC{\equiv}C)P_3N_3F_5$ appeared at 3.79 and 7.52 ppm respectively. A search of the literature indicated that the ^{31}P-NMR signals corresponding to the substituted phosphorus atom in the known alkene derived fluorophosphazenes, $N_3P_3F_5[C(OR)=CH_2]$ and $N_3P_3F_4[C(OR)=CH_2]_2$ (R = CH_3, C_2H_5), were observed in the range of 25.18 - 25.49 and 13.88 - 15.03 ppm, respectively [6]. For compound **3,** the spectral pattern was observed as two different triplet multiplets at 16.92 and 7.09 ppm but for compound **4** a simple triplet and a triple multiplet was obtained in the region 17.59 and 7.44 ppm. For the starting material $[(FcC{\equiv}C)_2PN](F_2PN)_2]$, two triplet multiplets were obtained at -24.83 and 6.36 ppm. So the chemical shift corresponding to substituted phosphorus atom undergoes a large shift (~34 ppm for the mono substituted compounds **1** and **2**, and ~ 42 ppm for geminal di substituted compounds **3** and **4** whereas the unsubstituted atom signal showed only a small change (<1 ppm). The ^{19}F-NMR spectrum for **1** and **2** showed almost a similar pattern of doublet multiplet but the signal corresponding to PF(C) changed from -49.23 ppm in compound **1** to -55.78 ppm in compound **2** whereas the same for the starting material came in the region -43.25 ppm. For **3** and **4** the changes of shift was very little (<1 ppm) between them and in comparison to the starting material also.

The -C≡C- stretch appeared as a weak band between 2260-2100 cm^{-1} for the starting materials mono and geminal bis(ethynylferrocene) derived fluorophosphazene but complete disappearance of such bands in all the products (**1-4**) and appearance of weak -C=C- bands in regions 1585-1638 cm^{-1} indicated the formation of alkenyl fluorophosphazene derivatives. While the identity of compounds **2** and **4** were further confirmed by high resolution mass spectral studies, the stereochemistry of the internal alkene units of **4** could not be ascertained due to complexity of the spectra and inability to get good crystals for X ray structural studies.

Conclusions

The first reactions exploring the reactivity of organozinc reagents with cyclophosphazene derived alkynes have been carried out using allylzinc bromide leading to the synthesis of a range of novel terminal alkene derived fluorinated cyclophosphazenes. The reactions of allylzinc bromide with mono and bis geminal ethynylferrocene derived fluorophosphazenes under different stoichiometries and reaction conditions have resulted in products having the allyl group regioselectively substituting on the alkyne unit with no substitution on the P-F bonds. While we have been able to add more than one allyl group on the unsymmetrical alkyne having both the electron withdrawing fluorophosphazene and electron releasing ferrocene groups, the preferred mode of addition of the allyl group has been on the alkynyl carbon adjacent to the ferrocene unit. The new compounds have been characterized by both spectral and structural studies. Interestingly, X-ray structural studies on the fluorophosphazene having geminally disusbtituted alkyne units indicated that one of the allyl group is attached in the *trans* manner to the double bond while the other allyl group is attached in the *cis* manner. quite likely that such reactions could be of value in preparing phosphazene-containing dendrimers The new fluorophosphazene derivatives having multiple terminal alkene units at its periphery are interesting precursors and provide scope for carrying out a variety of terminal alkene based reactions such as olefin metathesis, polymerization and alkene oxidation reactions. These reactions could also be of value in preparing new examples of phosphazene-containing dendrimers.

Experimental

General procedures

All manipulations were carried out using standard Schlenk techniques under a nitrogen atmosphere. THF was freshly distilled from sodium benzophenone ketyl and hexane was distilled and dried over sodium before use. Ethynylferrocene [15], hexafluorophosphazene [16], mono ethynylferrocene derived fluorophosphazene [9c] and geminal ethynylferrocene derived fluorophosphazene [9c] were prepared according to reported procedures. Hexa chlorophosphazene, allyl bromide and n-butyl lithium were procured from Aldrich and used as such.

Instrumentation

^{1}H and $^{13}C\{^{1}H\}$, $^{31}P\{^{1}H\}$, $^{19}F\{^{1}H\}$ spectra were recorded on a Bruker Spectrospin DPX-300 NMR spectrometer at 300 and 75.47, 121.48, 282.37 MHz, respectively. IR spectra in the range 4000-250 cm^{-1} were recorded on a Nicolet Protége 460 FT-IR spectrometer as KBr pellets. Elemental analyses were carried out on a Carlo Erba CHNSO 1108 elemental analyzer. Mass spectra were recorded on a Bruker Micro TOF QII quadrupole time-of-flight (Q-TOF) mass spectrometer.

X-ray crystal structure determination

Suitable crystals of compounds **1** and **3** were obtained by slow evaporation of their saturated solutions in ethyl acetate, hexane and toluene mixtures. Single-crystal diffraction studies were carried out on a Bruker SMART APEX CCD diffractometer with a Mo Ka (l ¼ 0.71073 Å) sealed tube. All crystal structures were solved by direct methods. The program SAINT (version 6.22) was used for integration of the intensity of reflections and scaling. The program SADABS was used for absorption correction. The crystal structures were solved and refined using the SHELXTL (version 6.12) package [17-20]. All hydrogen atoms were included in idealized positions, and a riding model was used. Non-hydrogen atoms were refined with anisotropic displacement parameters. Table 2 gives the data collection and structure solution parameters for compounds **1** and **3**.

Synthesis

Reaction of mono(ethynylferrocene) derived fluorophosphazene with allylzinc bromide in 1:6 molar ratio.

Activated zinc dust (0.18 g, 2.73 mmol) was taken in a 50 mL two necked round bottomed flask and to it allyl bromide (0.33 g, 2.73 mmol) was added along with 10 mL of dry THF. The mixture was stirred for 1h at melting ice temperature. Formation of allylzinc bromide was indicated by complete dissolution of zinc in the mixture. After bringing to room temperature , mono(ethynylferrocene) fluorophosphazene (0.20 g, 0.45 mmol) was added to it along with 5 mL of THF. The reaction mixture was refluxed under nitrogen for 48 h and the progress was monitored using TLC. After 48 h, it was quenched using a saturated solution of ammonium chloride over a period of 20 minutes. The reaction mixture was extracted with ethyl acetate (3x15mL) and the organic layer was dried over $MgSO_4$. After filtration and removal of solvent in vacuum, the crude product was purified by column chromatography over silica gel using distilled hexane and ethyl acetate (1%) as the eluent. The products were finally isolated using preparative TLC and the first compound obtained was an orange coloured semisolid **1** in (0.17 g,13%) yield and the second compound **2** was also obtained as an orange coloured viscous liquid in (0.31 g, 22%) yield. Compound **1** on keeping in hexane/ethyl acetate mixture for few days yielded dark red crystals of pure $[FcC(CH_2CH{=}CH_2){=}CH(F)PN](PNF_2)_2$ **(1).** Yield: (0.17 g, 13%). IR (KBr, ν/cm^{-1}): 2917s, 2849s, 2360w, 2171w, 1739w, 1637w, 1590w, 1462w, 1380w, 1268vs,1032m, 925m, 822m, 729w. ^{1}H NMR: (δ, 300 MHz, $CDCl_3$): 3.54 (2H, m, $CH_2CH=CH_2$), 4.07-4.48 (9H, m, Cp*H*), 5.04-5.14 (2H, d, $CH_2CH=CH_2$), 5.84 (1H, m, $CH_2CH=CH_2$), 5.88 (1H, m, C=C*H*). ^{13}C NMR: (δ, 300 MHz, $CDCl_3$**):** 35.88, 41.03 ($CH_2CH=CH_2$), 69.45, 69.85, 70.34, 70.42, (Cp*C*), 116.29, 116.51, (*C*H=*C*H_2), 136.08, 136.97 (*C*=*C*H). ^{31}P{^{1}H} NMR: δ 36.36 [dm (J_{P-F} = 1005.9 Hz) –PF(C)], 7.81 [t (J_{P-F} = 911.8 Hz), –PF_2]. ^{19}F{1H} δ -69.01 [dm, $^1J_{P–F}$ = 841.4 Hz, -PF2], -67 [dm, $^1J_{P–F}$ = 909.2 Hz, -PF2,], -49.23 [dm, $^1J_{P–F}$ = 910 Hz, -PF(C)]. HRMS calcd for [FcC($CH_2CH=CH_2$)=CH(F)PN](PNF_2)$_2$ (**1**) 481.9827 found 481.9837.

Characterization of [FcC($CH_2CH=CH_2$)=C($CH_2CH=CH_2$)(F)PN](PNF_2)$_2$ **(2)**.

Yield: (0.31 g, 22%). IR (KBr, ν/cm^{-1}): 2917s, 2848s, 2360w, 1637w, 1590m, 1461w, 1264vs, 1106w, 1056m, 922m, 821m, 729w, 688w. ^{1}H NMR: (δ, 300 MHz, $CDCl_3$): 3.13-3.53 (4H, m, $CH_2CH=CH_2$), 4.13-4.62 (9H, m, Cp*H*), 5.09 (4H, m, $CH_2CH=CH_2$), 5.83 (2H, m, $CH_2CH=CH_2$). ^{13}C NMR: (δ, 300 MHz, $CDCl_3$**):** 35.02, 42.20 ($CH_2CH=CH_2$), 69.63, 69.85, 70.35, 70.79, (Cp*C*), 116.49, 116.74, (*C*H=*C*H_2), 134.67, 135.06 (*C*=*C*). ^{31}P{^{1}H} NMR: δ 35.08 [dm (J_{P-F} = 1030.2 Hz) –PF(C)], 7.68 [t (J_{P-F} = 922.6 Hz), –PF_2]. ^{19}F{1H} δ -69.28 [dm, $^1J_{P–F}$ = 751 Hz, -PF2], -66.41 [dm, $^1J_{P–F}$ = 869 Hz, -PF2,], -55.78 [dm, $^1J_{P–F}$ = 999.6 Hz, -PF(C)]. HRMS calcd for compound [FcC($CH_2CH=CH_2$)=C($CH_2CH=CH_2$)(F)PN](PNF_2)$_2$ **(2)**: 521.0061, found: 521.0062.

Reaction of geminal bis(ethynylferrocene) derived fluorophosphazene with allylzinc bromide in 1:4 molar ratio.

Reaction of allyl bromide (0.37 g, 3.11 mmol) with activated zinc dust (0.18 g, 3.11 mmol) was carried out with geminal bis(ethynylferrocene) derived fluorophosphazene (0.48 g, 0.77 mmol) in 1:4 molar ratio and worked up and purified by adopting the same procedure for compound **1** and **2**. The product isolated as an orange coloured viscous liquid **3** in (0.23 g, 42%) yield which on keeping in hexane/ethyl acetate mixture yielded dark red crystals of pure $[FcC(CH_2CH=CH_2)=CHPN]_2(PNF_2)_2$ **(3)**. Yield: (0.23 g, 42%). Mp: 140-142 ˚C. IR (KBr, ν/cm^{-1}): 2955m, 2917s, 2849m, 2360w, 1638w, 1585m, 1461w, 1411w, 1267vs, 1241vs, 998w, 916m, 819m, 779m, 722w, 667w. ^{1}H NMR: (δ, 300MHz, $CDCl_3$): 3.59 (4H, m, CH_2CH=CH$_2$), 4.14-4.52 (18H, m, CpH), 5.10-5.19 (4H, m, CH$_2$CH=CH_2), 6.01-5.90 (2H, m, CH$_2$CH=CH$_2$), 6.07 (2H, m, C=CH). ^{13}C NMR: (δ, 300 MHz, $CDCl_3$**):** 37.05, 43.94 (CH$_2$CH=CH$_2$), 69.88, 69.95, 70.01, 71.30 (CpC), 116.66, 118.32, (CH=CH$_2$), 135.44, 135.99 (C=CH). ^{31}P{^{1}H} NMR: δ 16.92 [tm, ($^2J_{P-P}$ = 53.45 Hz) –P(R)$_2$], 7.09 [tm (J_{P-F} = 906.36 Hz), –PF$_2$]. ^{19}F{^{1}H} NMR: δ -67.42 [dm, F (J_{P-F} =999.6 Hz) –PF$_2$]. HRMS calcd for compound $[FcC(CH_2CH=CH_2)=CHPN]_2(PNF_2)_2$ **(3)**: 713.0287, found 713.0218.

Reaction of geminal bis(ethynylferrocene) derived fluorophosphazene with allylzinc bromide in 1:14 molar ratio.

Reaction of allyl bromide (2.01 g, 16.60 mmol) with activated zinc dust (1.09 g, 16.60 mmol) was carried out with geminal bis(ethynylferrocene) derived fluorophosphazene (0.75 g, 1.79 mmol) in 1:14 molar ratio and worked up and purified by adopting the same procedure for compound **1** and **2.** The products were isolated using preparative TLC and the first compound obtained was an orange coloured viscous liquid **3** in (0.27 g, 32%) yield and the second compound **4** was also obtained as an orange coloured viscous liquid in (0.21 g, 24%) yield. The compound **3** on keeping in hexane/ethyl acetate mixture yielded dark red crystals of pure compound $[FcC(CH_2CH=CH_2)=CHPN]_2(PNF_2)_2$.

Characterization of [{FcC($CH_2CH=CH_2$)=C($CH_2CH=CH_2$)}{CH=($CH_2CH=CH_2$)PN]$_2$(PNF_2)$_2$ **(4)**.

Yield: (0.21 g, 24%). IR (KBr, *ν*/cm^{-1}): 2917s, 2848s, 2360w, 1637w, 1588w, 1462w, 1240s, 1106m, 917m, 818m, 720w, 617m. ^{1}H NMR: (*δ*, 300 MHz, $CDCl_3$): 3.63-3.41 (6H, m, C*H*$_2$CH=CH$_2$), 4.16-4.55 (18H, m, Cp*H*), 5.21-4.90 (6H, m, CH$_2$CH=C*H*$_2$), 6.10-5.82 (3H, m, CH$_2$C*H*=CH$_2$), 6.10-5.82 (1H, m, C=C*H*). ^{13}C NMR: (*δ*, 300 MHz, $CDCl_3$**):** 32.08, 36.64, 36.74 (*C*H$_2$CH=CH$_2$), 69.93, 70.05, 70.09, 70.48 (Cp*C*), 116.44, 116.67, 118.32, 118.75 (*C*H=*C*H$_2$), 132.21, 135.57, 136.43 (*C*=*C*). ^{31}P{^{1}H} NMR: *δ* 17.59 [t, ($^2J_{P-P}$ = 50.60 Hz) –P(R)$_2$], 7.44 [tm ($^1J_{P-F}$ = 917.9Hz), –PF$_2$]. ^{19}F{^{1}H} NMR: δ -67.39 [dm, (J_{P-F} =931.8 Hz) –PF$_2$]. HRMS calcd. for compound [{FcC($CH_2CH=CH_2$)=C($CH_2CH=CH_2$)}{CH=($CH_2CH=CH_2$)PN]$_2$(PNF_2)$_2$ **(4)**: 754.0679, found 755.0754 [M+H]$^+$.

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Appendix A. Supplementary material

CCDC 1000576 and 1000577 contains the supplementary crystallographic data for the compounds discussed in this publication. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

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Highlights

- First reactions of organozinc reagents with cyclophosphazenes derived alkynes.
- Reaction takes place stereospecifically on alkynes and not on the P-F bonds.
- Alkyne derived phosphazenes were converted to terminal alkene derivative phosphazenes.
- Phosphazene derivatives having multiple terminal alkene units were synthesized.